

CURRENT POSITION	Assistant Professor <i>Cornell University</i> <i>Department of Statistics and Data Science</i>	2021 -
EDUCATION AND PREVIOUS ACADEMIC POSITIONS	Principal Researcher (Post-doc) <i>The University of Chicago, Booth School of Business</i>	2018 - 2021
	Ph.D. in Statistics <i>University of Washington</i>	2012 - 2018
	B.A. in Applied Math; Economics <i>Rice University</i>	2006 - 2010
STATISTICAL PUBLICATIONS	1) Tao, K., Wang, Y. S., Ruppert, D. “Bayesian Analysis of Regression Discontinuity Designs with Heterogeneous Treatment Effects” <i>Accepted at Journal of Causal Inference</i> 2) Wang, Y. S., Kolar, M., Drton, M. (2025). Confidence Sets for Causal Orderings. <i>Journal of the American Statistical Association</i>, In press. 3) Drton, M., Garrote-López, M., Nikov, N., Robeva, E., and Wang, Y. S. (2025). Causal discovery for linear non-Gaussian models with disjoint cycles. In Chiappa, S. and Magliacane, S., editors, <i>Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence</i>, volume 286 of <i>Proceedings of Machine Learning Research</i>, pages 1064–1083. PMLR 4) Zhao, B., Zhai, P. S., Wang, Y. S., and Kolar, M. (2024). High-dimensional functional graphical model structure learning via neighborhood selection approach. <i>Electronic Journal of Statistics</i>, 18(1):1042–1129 5) Duggan DiDominic, K. L., Shapleigh, J. P., Walter, M. T., Wang, Y. S., Reid, M. C., and Regan, J. M. (2024). Microbial diversity and gene abundance in denitrifying bioreactors: A comparison of the woodchip surface biofilm versus the interior wood matrix. <i>Journal of Environmental Quality</i>, 53(5):565–576 6) Wang, Y. S. and Drton, M. (2023). Causal discovery with unobserved confounding and non-Gaussian data. <i>Journal of Machine Learning Research</i>, 24(271):1–61 7) Wang, Y. S., Lee, C. J., West, J. D., Bergstrom, C. T., and Erosheva, E. A. (2023). Gender-based homophily in collaborations across a heterogeneous scholarly landscape. <i>Plos one</i>, 18(4):e0283106 8) Zhao, B., Wang, Y. S., and Kolar, M. (2022). Fudge: A method to estimate a functional differential graph in a high-dimensional setting. <i>Journal of Machine Learning Research</i>, 23(82):1–82 9) Wang, Y. S., Lee, S. K., Toulis, P., and Kolar, M. (2021). Robust inference for high-dimensional linear models via residual randomization. In Meila, M. and Zhang, T., editors, <i>Proceedings of the 38th International Conference on Machine Learning</i>, volume 139 of <i>Proceedings of Machine Learning Research</i>, pages 10805–10815. PMLR 10) Wang, Y. S. and Drton, M. (2020). High-dimensional causal discovery under non-Gaussianity. <i>Biometrika</i>, 107(1):41–59	

- 11) Zhao, B., Wang, Y. S., and Kolar, M. (2019). Direct estimation of differential functional graphical models. In *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, 8-14 December 2019, Vancouver, BC, Canada*, pages 2571–2581
- 12) Chen, W., Drton, M., and Wang, Y. S. (2019). On causal discovery with an equal-variance assumption. *Biometrika*, 106(4):973–980
- 13) Drton, M., Fox, C., and Wang, Y. S. (2019). Computation of maximum likelihood estimates in cyclic structural equation models. *The Annals of Statistics*, 47(2):663–690
- 14) Varshney, A., Wang, Y. S., Watson, R. A., Noll, A., Rossi, J., Shah, P. B., Kaneko, T., Giugliano, R. P., and Adler, D. S. (2018). Abstract 12195: Relationship between transcatheter aortic valve replacement wait time and mortality in patients with symptomatic severe aortic stenosis. *Circulation*, 138(Suppl.1):A12195–A12195
- 15) Chen, Y.-C., Wang, Y. S., and Erosheva, E. A. (2018). On the use of bootstrap with variational inference: Theory, interpretation, and a two-sample test example. *The Annals of Applied Statistics*, 12(2):846–876
- 16) Wang, Y. S., Matsueda, R. L., and Erosheva, E. A. (2017). A variational EM method for mixed membership models with multivariate rank data: An analysis of public policy preferences. *The Annals of Applied Statistics*, 11(3):1452–1480
- 17) Wang, Y. S. and Drton, M. (2017). Empirical likelihood for linear structural equation models with dependent errors. *Stat*, 6(1):434–447

AWARDS

College of Comp. and Info. Science Teaching and Advising Excellence Award (2025)
 Tom Ten Have Award, American Causal Inference Conference (2023)
The Tom Ten Have Award is awarded at each ACIC to two early career researcher poster presenters for posters that exhibit exceptionally creative or skillful research on causal inference

PRESENTATIONS

2026
 Isaac Newton Institute, Cambridge University

2025
 Binghamton University, Department of Statistics

2024
 University of Washington, Statistics and Machine Learning for the Social Sciences
 ASA Conference on Statistical Learning and Data Science
 University of Connecticut, Department of Statistics

2023
 TU Munich, Computation, Information and Technology
 University of Texas at Austin, Salem Center for Public Policy
 New England Statistics Symposium (NESS)
 Western North American Region (WNAR) of the International Biometric Society
 Algebraic Statistics for Ecol. and Biol. Systems, Inst. for Math. and Stat. Innovation
 Computational and Methodological Statistics (CMStat)

2022
 UPSTAT
 Online Causal Inference Seminar
 Johns Hopkins Causal Inference Working Group
 Cornell Center for Applied Math

2021

Colorado State, Statistics
Emory, Quantitative Theory and Methods
University of Southern California, School of Business
Cornell, Statistics and Data Science
University of Waterloo, Statistics and Actuarial Science
University of Toronto, Statistics
Stevens Institute of Technology, School of Business
Temple, School of Business

2020

Texas A&M, Statistics
Shiga University, International Symposium on Data Science Research and Practice

Prior to 2020

JSM (2015, 2017, 2018, 2019)
CMSTAT (2018)
American Sociological Association Annual Meeting (2018)
Wisconsin Institute for Discovery at UW-Madison (2017)
Toyota Technical Institute - Chicago (2017)

TEACHING

Cornell University
STSCI / INFO / ILRST 3900: Causal Inference
STSCI 6940: Graphical Models (Special Topics)
BTRY 6020: Statistical Methods II

University of Washington
STAT 311: Elements of Statistical Methods

STUDENTS SUPERVISED

PhD Thesis Advisor:
Nabil Boulos (ongoing, joint with Marty Wells)
Filippo Fiocchi (ongoing)
Sithija Manage (ongoing, joint with Marty Wells)
Shira Mingelgrin (ongoing)
Kevin Tao (ongoing, joint with David Ruppert)

PhD Thesis Committee Member:
Daniel Coulson (ongoing, Statistics)
Jiwoo Kim (ongoing, Psychology)
Daniel Molitor (ongoing, Info Sci)
Lauren San Diego (ongoing, Nutrition)
Samantha De Leòn Sautu (ongoing, Human Development)

Katie Duggan (2024, Bio and Envir Engineering)
Inbeom Lee (2024, Statistics)

Undergraduate Independent Study
Abby Sachar (2022-23)
Ziqing Wang (2021-22)

TECHNICAL REPORTS AND SOFTWARE

- 1) Wang, Y. S., Erosheva, E. A. (2016) “On the relationship between set-based and network-based measures of gender homophily in scholarly publications” [[arXiv](#)]
- 2) Wang, Y. S., Erosheva, E. A. (2015) “Fitting mixed membership models using `mixedmem`” [[CRAN](#)]

PROFESSIONAL
SERVICE

Journal Referee: Annals of Applied Statistics; Annals of Statistics; Bernoulli; Biometrika; Biometrics; Computational Statistics and Data Analysis; Electronic Journal of Statistics; IEEE Transactions on Neural Networks and Learning; International Journal of Approximate Reasoning; Journal of the American Statistical Association; Journal of Causal Inference; Journal of Computational and Graphical Statistics; Journal of Machine Learning Research; Journal of the Royal Statistical Society: Series B; Statistica Sinica

Conference Referee: AISTATS; ICML; NeurIPS; UAI; CLear

INDUSTRY
EXPERIENCE

Susquehanna International Group
Assistant Trader Intern

2013

Deloitte
Strategy and Operations Consultant

2010 – 2012